

Customers perspectives on adoption of cloud computing in banking sector

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Abstract The advent of cloud computing has transformed the role of the Internet in many businesses and organizations. Currently, banks are increasingly adopting cloud technologies to fulfil their varied purposes and to create a flexible and agile banking environment that can quickly respond to new business needs. However, past studies tend to focus more on the adoption issues of cloud computing from the organizational perspective with little attention paid on the users' view of these cloud-based services. Therefore, this paper attempts to investigate the factors influencing cloud computing adoption in the banking sector from the customers' perspective and to propose an adoption model for this purpose. The model is mainly developed based on the TAM-diffusion theory model (TAM-DTM) with the introduction of three new constructs namely trust, cost, and security and privacy. Questionnaires were randomly distributed to 162 bank customers in Malaysia. Survey data were analyzed using the partial least squares (PLS) method while SmartPLS was used to test the hypotheses and to validate the proposed model. The results suggest that trust, cost, and security and privacy can be successfully integrated within the TAM-TDM. The security and privacy constructs exhibited strong positive influence on perceived ease of use, perceived usefulness, and trust. The study concludes that perceived usefulness, perceived ease of use, cost, attitudes toward cloud and trust

significantly influence users' behavioral intention to adopt cloud computing. Thus, the finding of this study will enable banks to focus more on customer perspectives on cloud-based applications and identify their attitude towards their adoption.

Keywords Banking sector · Customer perspective · Cloud computing · TAM-diffusion theory model

1 Introduction

The banking industry is facing unprecedented changes. Nowadays control is under the hands of customers instead of the bank. The new business models are driving by customers. Their usage of technology besides changes in social and household dynamics is driving business transformation. Banks necessity to respond to this new customer driven environment with modernization in business models, procedures and information technology (IT). Banks will have a considerably capability to provide reliable service to customers over the branches, geographies and also integrate a surplus of dislocate customer information and analytics [11]. Banks necessity to look at the things from the outside, from the consumer's point of view, to change something in the banking system. The dramatic changes happening in banking need new ways to enlarge profitability and revenues [63]. Cloud technology offers secure deployment options that can assist banks improve new customer experiences, facilitate impressive collaboration and develop rapidity to market all-while increasing IT efficiency. There has been growing interest in the use of cloud computing in multiple industries [45, 78, 90], and banking is no exception.

Cloud computing supports banks to make new markets and services to distinguish from competition and develop

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the ways customers' access and use the bank's products and services. Banks that take benefit of cloud computing are well positioned to answer to financial uncertainties, interconnected universal financial systems and demanding customers. They are able to use information to improve customer segmentation methods and to improve more centralized services that are aligned with customer requests. Banks similarly can enhance their channel investments and distinguish themselves through customer service excellence [63, 102].

The pressure of today's competitive marketplace force many organizations, including banks, to rapidly deploy innovation in their current business models [62]. Cloud computing in the banking context consists of two different types of users [68]. The first category of users is the banks that benefit and deploy cloud services in their current business model. The second are customers who perform their daily financial transactions using the cloud-based applications provided by the banks [62, 68]. The use of cloud computing enables banks to create a flexible and agile banking environment that can quickly respond to new business needs [102]. Furthermore, cloud technology offers secure deployment options that can help banks develop new customer experiences, facilitate effective collaborations and improve speedy access to markets, while increasing IT efficiency. This gives employees the advantage of focusing precisely on the bank's vision and mission [20, 63].

On the other hand, according to Benton [20], there are two areas in which cloud-based applications can be used to assist bank customers. First, they provide a vast opportunity for customers to dramatically increase their interactions with the banks [21]. Second, cloud-based applications, specifically software as a service (SaaS), can help bank customers fulfil their daily financial transactions more easily and without any extra cost [62] as against traditional client server technologies that impose huge cost, time, and resource barriers [80]. However, with cloud-based applications, customers do not need to install software [91].

Therefore, it is of interest to further understanding the factors that foster cloud computing adoption in the banking industry [89]. Theoretically, the adoption of cloud-based applications by bank customers have been shown to create a wide range of opportunities for them [62]. In practice however, most bank customers still have cold feet towards the adoption of banks' cloud based applications as they still prefer to use the existing client/servers solutions offered by these banks [20]. According to Catteddu and Hogben [27], barriers such as data privacy and data security are the major hindrance of SaaS adoption. Banks are categorized as a very high risk level and sensitive organizations [66]. Therefore, primary concerns such as data privacy and data security are the big migration for bank customers to cloud

services [140]. This trend dates back to the history of Internet banking, where most bank customers thought the Internet was the most dangerous place to perform their daily financial transactions and therefore avoided its use [144]. However, the numerous and tempting benefits of online banking have continued to lure bank customers, most of who now see it as not only convenient but secure and has dramatically boosted the number of Internet banking users in recent times [70]. Similarly, cloud-based applications in the banking context are currently in their early stage [20] and has been predicted to take around ten to fifteen years to convince customers to make this trend [21]. Thus, due to the novelty of cloud-based applications in the banking context, the decision making in adoption of cloud services has not been empirically addressed [88] and as a result, there is a lack of frameworks for adoption of cloud-based applications by bank customers [62]. Extant literature shows that there is a lack of study on customers as the end users of cloud computing in the banking context [62, 68]. Most prior research on the cloud computing adoption has only focused on banks rather than customers [62]. Hence, there is a fundamental need for banks to focus more on customers' perspectives on cloud-based applications so as to identify their attitude towards adoption [20, 68].

Accordingly, this study will identify the factors influencing adoption of cloud computing in the banking sector from the customers' perspectives. Specifically, it will highlight the importance of security/privacy, cost, and trust issues as well as develop a research model integrating these factors; and empirically analyse the model using structural equation modelling (SEM) with data from banking users in Malaysia. This is aimed to address the primary research question of this study:

- What are the factors which influence the adoption of cloud-based applications by bank customers?

The rest of the paper is therefore structured as follows: Sect. 2 presents a review of related work on cloud computing, Cloud computing in banking sector, and Sect. 3 discusses the overview of theory used for this study, Sect. 4 presents the research methodology, Sect. 5 and 6 look at data analysis and discussion of findings, and Sect. 7 conclusion of the study. Section 8 is about contribution of the study. The last part of the paper is limitations and future work.

2 Literature review

2.1 Cloud computing

The term cloud computing emerged in the fourth quarter of 2007 on a collaborative project between IBM and Google

[138, 147]. The National Institute of Standards and Technology (NIST) defines cloud computing as: “a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction” [95]. Some of its advantages include on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service for business [95]. Buyya et al. [26] defined cloud computing as: “Cloud is a parallel and distributed computing system consisting of a collection of inter-connected and virtualized computers that are dynamically provisioned and presented as one or more unified computing resources based on service-level agreements (SLA) established through negotiation between the service provider and consumers”. Although several definitions of cloud computing abound, some cloud characteristics are notably common including: (1) self-service interface; (2) elastic capacity and the illusion of infinite resources; (3) pay-per-use (no on-going commitment, utility prices); and (4) virtualized or abstracted resources [10].

2.2 Cloud computing in banking sector

Cloud computing can be described as an offspring of the new IT age. Presently, many banking sectors employ cloud computing technologies to fulfil their different purposes. By using cloud technologies, banks can create a nimble and flexible banking environment that is able to respond quickly to new business requirement. This is also cost saving as financial institutions need only pay only for functional consumption and the services they use [102].

Although many large banks are confident about the acceptance of cloud technology, others are more careful and await the security and regulatory challenges to be addressed before jumping onto the bandwagon. Based on a 2010 survey of bank customers interest in 186 banking technologies conducted by Bank Systems Technology and Information Week Analytics, majority (73%) of the candidates attributed their interest to the ability of cloud services to meet customer request rapidly as well as to have a scale in the cloud. However, it is expected for banks to lead the world in cloud adoption and nobility of SaaS solutions. With serve and tight monetary restrictions, mature monetary software marketplace and rapidly changing regulations, there is need for banks to pursue inventive solutions to meeting business requirements in today server challenging environment [7].

However, the banking sector still harbours considerable concerns with regards to cloud technologies. Security, privacy, location of data, data integrity, confidentiality,

availability and user trust are the most prominent among these concerns [22, 109].

Security, trust, and privacy always remain challenges for organizations that adopt cloud computing [28]. According to Bose et al. [22], security, privacy and trust plays a paramount and central role in both cloud computing and banking services. To some extent, cloud security is similar to security controls in online banking systems. They both need to incorporate technological means, such as firewalls, intrusion detection/prevention systems, anti-virus, authentication, authorization and encryption, etc. to authenticate business information in different settings. Trust can be defined as “the intention to accept vulnerability based upon positive expectations of the intentions or behavior of another” [111]. Trust is an significant factor affecting adoption of new technology. Trust in a cloud environment determined by the privacy and security properties of the selected deployment model. If cloud service providers follow strict security and privacy policies, trust may be improved, which in turn positively affects customers’ intention toward using these services [9]. In this study perceived security and perceived privacy are directly related to trust in cloud storage services and indirectly related to behavioral intentions toward adopt cloud computing.

As most of the cloud services are executed on virtual machines, the service providers should emphasis on securing a holistic virtual environment and maintaining several different layers of protection to increase customers’ trust in cloud computing. The most critical security and trust issues fall under such categories as: access control, confidentiality, integrity, reliability, availability, recoverability, accountability, and long-term viability [22].

Clients must trust the cloud provider the same way they would be willing to trust banks to put their money into them. Similarly, the cloud providers must demonstrate that they are reliable and trustworthy. Therefore, for wide-spread adoption of cloud computing, we contend that customers should be able to store their data in the cloud with the same confidence that they store their money and other valuables in banks today.

The bank is trustful not just for its safety, but also for its assurance/commitment. Users can easily deposit, access, transfer, and withdraw their money on demand. Even if the bank encounters threats, users are guaranteed to receive their money back. Comparatively, less trust is put on cloud because it is yet to provide users with sufficient security and assurance. For instance, Using cloud, banks can hide some of the intricacies of their operations from end users, which can help attract a broader range of consumers. Because the complexity is not obvious to the customer, banks can expand their product and service sophistication without also increasing the level of end-user knowledge

necessary to use or maintain the product or service. For example, upgrades and maintenance can be done in the background without the customer having to participate [63].

Cloud users lack control over their data; they do not know the exact storage location these data. These unclear security challenges remain the primary reason for this mistrust. Customers are ultimately responsible for the security and integrity of their own data, even when it is held by a service provider [109]. Therefore, in order to gain consumers' trust, cloud providers must offer better transparency and more consumer control of data and processes [22].

Another major issue for cloud adoption is the security and privacy of users' data used for banking applications [109]. Lastly, the issue of commitment must also be taken very seriously given that commitment is a key variable for establishing and maintaining successful long-term relationships with customers [98, 133]. This has made commitment development a major objective for managers to establish stable relationships and networks with existing customers [14].

2.3 Related works on cloud adoption

Although several studies have been conducted by various researchers in the domain of cloud computing adoption in different context, the literature review findings reveal that majority of these studies were mainly focused on cloud computing adoption at the organizational level and the implementation of cloud computing in terms of its technical aspects, as shown in Table 1 [21, 24, 52, 73, 82, 91, 96, 103, 105, 109, 123, 125, 127, 140]. Khajeh-Hosseini et al. [71] assert that there are many factors that can influence the adoption of new technologies. These factors vary from focus on technology itself to user's attitudes and characteristics. Haider [54] highlighted the significant importance of customers' adoption decision in cloud computing. In fact, consumers are important stakeholders in cloud computing, as many well-known cloud solutions such as Microsoft SkyDrive (hosting Office Web App) and Google Drive (formerly Google Docs) target not only organizations but also individuals, and the latter is a critical factor in determining the popularity and success of these solutions. Therefore, it is necessary to expand knowledge on this group of users and to understand the factors that influence their adoption decisions [54]. Although Park and Ryoo [101] posit that there are few studies that have looked into user adoption behavior from previous information systems (IS) services to cloud computing, little is known about how these individuals adopt cloud computing for personal usage [84, 91, 124]. Subsequently, empirical studies in this area only examined a handful of antecedents

to individual adoption decision, such as perceived usefulness of cloud computing, perceived ease of use, privacy concerns, and access to software [81]. Many other factors that may influence individual adoption decisions have not been adequately analysed.

In addition, the majority of cloud adoption literature discussed security, privacy and cost as related issues and critical features that influence the adoption of cloud computing. Benton and Negm [21] claimed that a major advantage of the cloud, besides the limited period of time, is less costs. As an example, Accenture predicted that relocation of majority of existing applications to clouds infrastructure will save IT companies about 50% of their hosting related expenses yearly. Wu et al. [140] indicated that cloud computing can provide services on demand at lower cost with greater scalability. Rani and Gangal [109] noted that the unlimited infrastructure provided by cloud computing enables data storage and retrieval across multiple locations which can lead to data redundancy and data modification by unauthorized users. This often results in data loss, thus emphasizing the importance of privacy and data security in cloud computing applications used in the banking sector. This is in agreement with the assertion of Pearson [103] that security is the most challenging issue for cloud users. This security challenge, has been identified by majority of chief information officers (CIOs) in big organizations, as the main cause for the low acceptance rate of cloud computing by most users, thereby denying these users the benefits and opportunities offered by cloud computing. Also, Misra and Mondal [96] asserted that one of the most apparent shortcomings of cloud computing that prevents it from being a runaway success, is the concern for the security of data governance in the cloud environment. According to Marston et al. [91], "Cloud computing raises new privacy issues that require clear standards for custodians of this information who receive government requests for access to that information". Hence, loss of privacy is of particular concern to users when sensitive data is processed in the cloud [103].

Based on the above review, this study proposes three critical factors for cloud computing adoption. Also, the literature review findings suggest that security plays a critical role in the adoption of cloud computing [82, 99, 127]. Security issues also influence cloud users choices or methods of buying or paying for goods/services. Some of these cloud computing benefits and opportunities become more glaring especially when differentiated with a very large and crowded marketplace, reemphasizing the need for improved security standards by cloud providers [91]. A summary of the review findings is presented in Table 2.

According Table 2, it is obvious that security, privacy, and cost are predominant factors influencing cloud

Table 1 Findings of prior studies on cloud computing adoption

Author(s)	Context area	Factors affecting CC adoption
Lian [82]	E-government	Effort expectation, social influence, trust, perceived risk, security concerns, behavioral intention
Tehrani and Shirazi [127]	Medium size enterprises (SEs)	Relative advantage, complexity, privacy, security, cost, external support, decision maker's innovativeness, decision maker's cloud knowledge, employee's cloud knowledge, information intensity
Oliveira et al. [100]	Organization	Security concerns, cost savings Relative advantage, complexity Compatibility, technology readiness Top management support, firm size Competitive pressure, regulatory support
Burda and Teuteberg [25]	End-user adoption	Trust, risk, perceived usefulness, perceived ease of use, satisfaction, familiarity, reputation, intention to use
Stieninger and Nedbal [120]	SMEs	Individual factors (perceived usefulness, perceived ease of use, security, trust)
Alotaibi [6]	Users' perspective	Trust, anxiety, perceived risk, perceived ease of use, perceived usefulness
Stieninger et al. [121]	Organization	Compatibility, relative advantage, complexity, image, security, trust
Gupta et al. [52]	Small and medium Business	Cost savings, ease of use, reliability, security, privacy, sharing and collaboration, usage and adoption of cloud
Swamy [125]	Organization	Cost savings, security concerns, compliance, issues of management, control over the data
Pearson [103]	Non-government organizations	Privacy, security, trust, availability, compliance
Ogunde and Mehnen [99]	End-user's perspective	Security, cost, availability, compliance, perceived usefulness
Rani and Gangal [109]	Banking	Security, privacy, confidentiality, data integrity, availability, recoverability
Piplode and Singh [105]	Organization	Privacy, security, performance, availability
Udoh [131]	Organization	Perceived usefulness, perceived ease of use, attitude, trust
Wu [139]	Government organizations	Security, trust, cost
Brohi and Bamiah [24]	Government organizations	Privacy, interoperability, reliability, cost reduction
Marston et al. [91]	Business organizations	Data privacy, data access
Sultan [123]	Education	Performance, latency, security, privacy, reliability, cost
Misra and Mondal [96]	Business organizations	Cost saving, agility/flexibility, scalability
Benton [20]	Banking	Cost saving, security, data privacy
Kim et al. [73]	Organization	Availability, security, performance, compliance, integration, cost

computing adoption. We shall therefore attempt to use these aforementioned factors as a guide in developing the proposed model for this study. Hence, the study shall aim to achieve the following objective: follows:

- To identify the factors which influence the adoption of cloud based applications by bank customers.

Looking at Table 1, it is concluded that most of the previous researches principally focus on the overall concept of cloud computing, such as positive elements of use, the advantages of using cloud computing and cloud

computing adoption. Furthermore, there have been few theoretical models proposed to understand the user perspectives for adoption of cloud-based applications banking sector. These studies illustrates the initial factors for cloud computing adoption in different context. In details, these factors are the summary of the most frequent adoption factors that have been used in the literature. These initial factors are used as a guideline for the next step of the research; research model development. This study elicited these preliminary determinants from the literature to develop the initial research model.

Table 2 Most important factors on cloud computing adoption

Factors	Authors										
	Kim et al. [73]	Benton [21]	Misra and Mondal [96]	Sultan [123]	Marston et al. [91]	Brohi and Bamiah [24]	Wu [139]	Piplode and Singh [105]	Udoh [131]	Rani and Gangal [109]	Ogunde and Mehnen [99]
Availability	●							●		●	●
Security/privacy	●	●		●	●	●	●	●		●	●
Performance	●			●				●			
Compliance	●										●
Integration	●									●	
Cost	●	●	●	●		●	●				●
Agility/ flexibility			●								
Scalability			●								
Latency				●							
Reliability				●		●					
Data access					●						
Interoperability						●					
Trust							●		●		
Confidentiality										●	
Recoverability										●	
Perceived usefulness									●		●
Perceived ease of use									●		
Control over data											
Sharing and collaboration											
Usage											
Relative advantage											
Complexity											
Compatibility											
Technology readiness											
Top management support											
Firm size											
Competitive pressure											
Regulatory support											
External Support											
Decision maker's innovativeness											
Decision maker's cloud knowledge											
Employee's cloud knowledge											

Table 2 continued

Factors	Authors										
	Kim et al. [73]	Benton [21]	Misra and Mondal [96]	Sultan [123]	Marston et al. [91]	Brohi and Bamiah [24]	Wu [139]	Piplode and Singh [105]	Udoh [131]	Rani and Gangal [109]	Ogunde and Mehnen [99]
Information intensity											
Effort expectation											
Social influence											
Perceived risk											
Behavioral intention											
Attitude									●		
Anxiety											
Satisfaction											
Image											
Reputation											
Issues of management											
Intention to use											
Factors	Authors										
	Pearson [103]	Swamy [125]	Gupta et al. [52]	Burda and Teuteberg [25]	Stieninger and Nedbal [120]	Stieninger et al. [121]	Oliveira et al. [100]	Alotaibi [6]	Tehrani and Shirazi [127]	Lian [82]	
Availability	●										
Security/privacy	●	●	●		●	●	●		●	●	
Performance											
Compliance	●	●									
Integration											
Cost		●	●				●		●		
Agility/flexibility											
Scalability											
Latency											
Reliability			●								
Data access											
Interoperability											
Trust	●			●	●	●		●		●	
Confidentiality											
Recoverability											
Perceived usefulness				●	●			●			
Perceived ease of use			●	●	●			●			
Control over data		●									
Sharing and collaboration			●								
Usage			●								
Relative advantage						●	●		●		
Complexity						●	●		●		
Compatibility						●	●				

Table 2 continued

Factors	Authors									
	Pearson [103]	Swamy [125]	Gupta et al. [52]	Burda and Teuteberg [25]	Stieninger and Nedbal [120]	Stieninger et al. [121]	Oliveira et al. [100]	Alotaibi [6]	Tehrani and Shirazi [127]	Lian [82]
Technology readiness							●			
Top management support							●			
Firm size							●			
Competitive pressure							●			
Regulatory support							●			
External support									●	
Decision maker's innovativeness									●	
Decision maker's cloud knowledge									●	
Employee's cloud knowledge									●	
Information intensity									●	
Effort expectation										●
Social influence										●
Perceived risk				●				●		●
Behavioral intention										●
Attitude										
Anxiety								●		
Satisfaction				●						
Image						●				
Reputation				●						
Issues of management		●								
Intention to use				●						

According Table 2, all study attempt to investigate the critical factors that will affect the decision to adopt cloud computing technology. More than 8 studies indicated that, the four most critical factors are: in sequence security, privacy, cost and trust. However, reviewing the previous research literature on cloud computing adoption in is worth consideration in order to understand the usage of cloud based applications and to identify the critical factors and basic theories used in these studies. Therefore, Table 2 summarized prior research on cloud computing adoption using cloud-based applications in different context.

3 Overview of theory used for proposed model

The technology acceptance model (TAM) is one of the best known models was deliberated by Davis [39] to predict and explain the acceptance and use of an Information System (IS), based on behavioural intention, and actual system usage. And is broadly used amongst numerous models and theories in the context of IS research [41] because of its simplicity and comprehensiveness [74].

The larger number of adoption studies used TAM and utilizes causal-explanatory statistical analysis for

Table 3 Descriptive statistics of respondents' characteristics

Measure	Value	Frequency (%)
Gender	Male	76 (46.92)
	Female	86 (53.08)
Age	<26	23 (14.2)
	26–34	106 (65.4)
	>35	33 (20.4)
Job	Student	136 (83.96)
	Office workers	11 (6.79)
	Expert	3 (1.85)
	Others	12 (7.40)
Education	Bachelor degree	43 (26.54)
	Master degree	56 (34.57)
	PhD degree or above	63 (38.89)
Experience using online banking services	<1 year	29 (17.90)
	1–2	36 (22.22)
	2–3	43 (26.54)
	More than 3	54 (33.33)

understanding relationships [16, 83, 100, 101, 115, 141]. Although TAM model has been used widely to study different environments, characteristics that are specific to cloud computing are not captured by this model. The problems using this approach are original TAM by Davis [39] utilizes only two constructs perceived ease of use and perceived usefulness, which have their own limitations in explaining the intention to use [136]; therefore, a number of researchers have pointed out the need to extend TAM by augmenting the original TAM with additional variables for better modeling of the intention to use. Next, information systems researchers have often highlighted the importance of predictive analytics instead of causal-explanatory statistical analysis that enables enhanced theory building and testing [34, 115, 119]. Also, there is a lack of considering the notion of perceived outcomes when forecasting behavior in this theoretical model [36].

Another theory which has received similar attention by scholars in explaining consumer behavior towards new technology is the Rogers' Diffusion theory. This study integrate two leading models (TAM and DOI) in order to explain user's intention to adopt cloud computing.

The general information provided by the TAM model alone is not adequate to motivate the consumer's for cloud in banking. Therefore in the present study an attempt has been made to integrate TAM and DOI models and an extended model. First, TAM and DOI models identify the perceived attributes of an innovation as key predictors explaining adoption. Second, both models feature users' intentions to adopt a technology (or their actual adoption/use) as their dependent variable. Third, both models apply most readily to situations where the individual user can voluntarily choose whether to adopt the innovation or not [46].

Besides the mainstream modification of technology acceptance model (TAM) modifications, there is a need to identify variations in some of the TAM versions. Studies such as [87, 129] have all attempted to use a combination of innovation diffusion theory (IDT) and TAM to develop integrated models [110], arguing that all versions of TAM were tautological, incomplete and too parsimonious. López-Nicolás et al. [87] developed a novel TAM-diffusion theory model (TAM-DTM) that was somewhat different from the traditional modifications of TAM. Although TAM-DTM shares some important features with the traditional TAM, it nevertheless possesses some unique features that distinguish it from other TAM modifications. These include its practicability and ease of use without the inclusion of moderating variables. Also, it employs the media influence (MI) as an initial construct that has direct impact on the social influence (SI) construct. In other words, the antecedent of SI in TAM-DTM is MI whereas for TAM3 and unified theory of acceptance and use of technology (UTAUT) model, SI has no antecedents. Thus, it can be argued that the effect of mass media during the developmental or early phases of innovation adoption are very dominant based on the TAM-DTM model. Furthermore, the influence on SI in the TAM-DTM model is felt across almost all of the constructs such as behavior intention (BI), perceived flexibility benefits (PFB), perceived ease of use (PEOU), and attitude toward mobile innovations (ATT), perceived usefulness (PU) and perceived status benefits. Lastly, the perception of TAM-DTM as media-oriented model for end-user technology acceptance in the context of a marketplace, distinguishes it from UTAUT and TAM that focus mainly on employee technology acceptance in the workplace context [139].

4 Research methodology

This research adopts purposive sampling approach and chooses Malaysian banking users, which already using online banking services. The study adopted a survey-based methodology. This method has been described as the most appropriate approach to adopt when trying to generalize the opinion of a sample group or to draw conclusions on a population based on a sample drawn from that population [100]. Since a diverse population in Malaysia uses online banking services, the respondents were selected from diverse groups of individuals consisting of workers from the private and public, domain experts, students and university campuses. They were asked to participate in a survey, if they had experience using online banking. Online banking services such as mobile applications (offering checking balance, order new chequebooks or stop payment orders in banking transactions) described for online banking customers. A total of 291 questionnaires were distributed, of which 162 usable questionnaires were collected indicating a response rate of 55.67%. The survey was conducted over a period of 2 months in February and March 2013. The gender distribution shows 47% male and 53% female respondents, while the educational background variable reveals that 63% of the respondents had earned a higher education degree, and about 82% had over 1 year of online banking experience. Detailed descriptive statistics relating to the respondents' characteristics are shown in Table 3.

The response bias was tested using independent samples of *t* tests based on the collected data in order to overcome the bias concerns common with the survey design methods [23]. The early and late respondent groups manifested response/non-response differences [37, 85, 130].

Late responses were characterized as those that resulted from the second reminder and follow-up efforts. Specifically, 98 responses were identified as early responses, and 64 were late. The mean values of the key study variables between early and late respondents were compared, and there was no significant difference between the two at $p < .05$ [76, 130]. The test shows no differences among means of early and late respondents for the variables of this study. The respondents were qualified individuals, indicating a good quality of data. The common method bias was examined using Harman's one-factor test [106]. Thus, the results indicate no significant common method bias was found in the data set [100].

As discussed earlier (see Table 2), factors such as trust, cost, security and privacy have a considerable effect on cloud computing adoption. Therefore, any effort to conduct a study on cloud computing adoption application must to consider the integration of these three major constructs

(trust, cost, security and privacy) into the TAM-DTM model. The extended version of TAM, known as TAM-DTM, has been widely used in previous studies on the adoption of new technologies [87, 139]. Hence, these three constructs were examined in a TAM model. Consequently, this research propose TAM model that combines related theories with imperative constructs such as trust, cost, security and privacy in order to gain more insight into the adoption of cloud computing based applications. Partial least squares (PLS) is suitable for sample sizes over 80, or when the sample is larger than ten times the number of independent variables [12]. This study has eight independent variables (e.g., the minimum required sample is 80). Therefore, in order to ensure sample availability, the minimum sample size required for this study was 80.

A pilot study was first performed in order to validate the preliminary questionnaire used for the study. According to Fornell and Larcker [43], Composite Reliability (CR) value is acceptable when it is higher than 0.7 while the Average Variance Extracted (AVE) value should be higher than 0.5. Based on the reliability test performed, all the indicators of the original model (ATT, BI, COST, PB, PEOU, PU, S&P, SI and T) had loading higher than 0.6 [56]. Except for all 3 items of Media Influence; MI1, MI2, MI3. Furthermore, the 3 items (MI1, MI2, MI3) of the Media Influence construct were unable to meet the acceptable Cronbach's Alpha, Composite reliability and AVE values. Thus, since removing these items resulted in the improvement of the items reliability and overall reliability of the relevant constructs, Media Influence construct was dropped from the measurement model.

4.1 Measurement

Scales were used to measure the constructs in the questionnaire used for data collection based on the research model. The measurements for social influence, perceived usefulness, perceived benefits and attitude constructs were adapted from López-Nicolás et al. [87] and Wu [139] studies. The measurements for the cost construct were adapted from studies conducted by Medyawati et al. [94], while the measurements for the security and privacy constructs were adapted from Shin [117] and Wu [139] respectively. The measurements for the trust construct were adapted from studies conducted by Kim et al. [72] and Wu [139] respectively. The measurement for the construct behavior intention were adopted from studies conducted by López-Nicolás et al. [87], Shin et al. [118], and Yen et al. [142]. A total of 33 items were derived from original measurements which had been applied in various research areas and models. This helped to enhance the validity of the questionnaire [1]. The items were then modified

Table 4 Constructs and measuring items

Construct	Definition of constructs	Measuring items	References
Social influence (SI)	The degree to which an individual perceived that important others believe he or she should use the cloud computing	SI1: Mass media affects me for using the cloud based application in banking transaction SI2: Expert opinions affect me for using cloud based application in banking transaction SI3: Oral communication affects me for using the cloud based application in banking transaction	López-Nicolás et al. [87], Shin [117], Wu et al. [140] and Venkatesh et al. [136]
Perceived benefits (PB)	Is defined as beliefs about improved conditions or gains brought about by a given recreational activity	PB1: Using the Cloud based apps solution is a representation of fashion in banking transaction PB2: Using the Cloud based apps solution is a symbol of status in banking transaction PB3: Using the Cloud based apps solution is a way to increase confidence in banking transaction PB4: Using the Cloud based apps solution makes me cheerful in my banking transaction	López-Nicolás et al. [87], Wu et al. [140] and Ajzen and Driver [5]
Attitude (ATT)	Refers to individuals' positive or negative evaluation of the behavior	ATT1: I am interested in the news about using the cloud based application in banking transaction ATT2: I enjoy trying new technology and services for my banking transaction ATT3: I want to be a pioneer to try using the cloud based application in my banking transaction	López-Nicolás et al. [87], Shin [117], Wu et al. [140] and Ajzen and Fishbein [3]
Perceived usefulness (PU)	The degree to which a person believes that using a particular system would enhance his or her job performance	PU1: Using the cloud based application enable me to do online transactions faster PU2: Using the cloud based application improves my online financial transaction performance PU3: The useful function of the cloud based application satisfies me	López-Nicolás et al. [87], Shin [117], Wu et al. [140], Davis [39] and Cheng et al. [31]
Perceived ease of use (PEOU)	The degree to which a person believes that using a particular system would be free of effort	PEOU1: The use interface of the cloud based application is friendly PEOU2: The procedure of using the cloud based apps solution is understandable PEOU3: It is easy to learn using the cloud based apps solutions PEOU4: It is easy to make use the cloud based apps solutions	López-Nicolás et al. [87], Shin [117], Wu et al. [140], Davis [39] and Cheng et al. [31]
Cost	Cost reduction (economical) in terms of data storage, subscription, low upfront cost (capital expenditure), cost control via scalability, elasticity of resources (scale up and down in a very short time)	Cost1: Cost is an important factor for me to use cloud applications Cost2: Financial transactions are more easily in the cloud without any extra cost Cost3: Customers can save cost and managing time with aids of cloud applications	Medyawati et al. [94], Ajzen and Driver [5] and Gupta et al. [52].
Security	Cloud computing security refers to the set of procedures, processes and standards designed to provide information security assurance in a cloud computing environment	S&P1: Using the cloud based apps can be secure S&P2: The cloud based application keeps clients information's private and confidential	Shin [117], Wu et al. [140] and Lallmahamood [77] (technopedia.com 2010)

Table 4 continued

Construct	Definition of constructs	Measuring items	References
Privacy	The rights and obligations of individuals and organizations with respect to the collection, use, retention, and disclosure of personal information	S&P3: I believe that cloud applications in bank are secure S&P4: My personal data keep safe in cloud computing applications	Shin [117], Wu et al. [140], Lallmahamood [77] and Chen and Zhao [30]
Trust	The willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the truster, irrespective of the ability to monitor or control that other party	T1: I trust the cloud because its vendors are competent and professional T2: The cloud vendor is one that keeps promises and commitments T3: The cloud vendor is one that keeps promises and commitments T4: I trust the cloud vendor because they keep my best interests in mind T5: Overall, the cloud is trustworthy	Kim et al. [72], Wu et al. [140] and Mayer et al. [93]
Behavioral intention (BI)	Is defined as the cognitive representation of a person's readiness to perform a given behavior and considered the immediate antecedent of behavior	BI1: Overall, I intend to use the cloud based apps solution in the future BI2: If possible, I will try to use cloud applications BI3: I plan to use cloud applications in the future BI4: Assuming cloud based app would be available on my job, I predict that I will use it on a regular basis in the future	López-Nicolás et al. [87], Shin [117], Wu et al. [140], Yen et al. [142] and Ajzen [4]

specially for the context of cloud computing. All the items in the questionnaire were measured using a five-point Likert scale, with responses ranging from “strongly disagree” to “strongly agree.” All the items used in the questionnaire are presented in Table 4.

4.2 Research hypothesis

As shown in Fig. 1, the proposed model is mainly based on the TAM-DTM [87], but with the addition of new constructs; Cost, Trust, Security and Privacy. The research model includes eight constructs: (1) SI, PB, ATT, PU, PEOU, and BI are adopted from TAM-DTM [87]. Based on these constructs, the research hypotheses for this study were formulated as follows:

SI is the degree to which individuals consider whether or not to use cloud computing because of the effect of mass media and expert opinions, as well as word-of-mouth.

Given that society has a strong influence on individual decision-making, be it in a family, peer group or business environment, our behaviours, attitudes and perceptions are influenced by stimuli and information from our social surroundings. Social influence in the form of, expert opinions and word of mouth affects the determinants of adoption of cloud computing services. The influence that society wields may have the ability to encourage

technology laggards to experiment with new technologies and change perceptions around the perceived benefits of cloud computing services.

According to López-Nicolás et al. [87], SI has positive effects on many constructs such as: ATT, PB, PU, and PEOU. Therefore, we assume that SI also has a positive effect on ATT, PB, PU, and PEOU.

H1 SI has a positive effect on ATT.

H2 SI has a positive effect on PB.

H3 SI has a positive effect on PEOU.

H4 SI has a positive effect on PU.

The user acceptance of new Information technologies or services is determined by both functional and non-functional reasons [104]. Referring to López-Nicolás et al. [87], the acceptance of new ITs or services can be viewed as the representation of advanced social status. It is apparent that the consideration of social status must be based on non-functional concerns. That is, PB emphasizes the non-functional concerns while PU highlights functional concerns. It is apparent, furthermore, that PB is the degree to which individuals perceive using the SaaS as a representation of fashionability, a symbol of status, and a way to increase confidence. PU is the degree to which individuals perceive that the use of the SaaS is helpful to efforts to do things faster and better. Additionally, ATT is the degree to

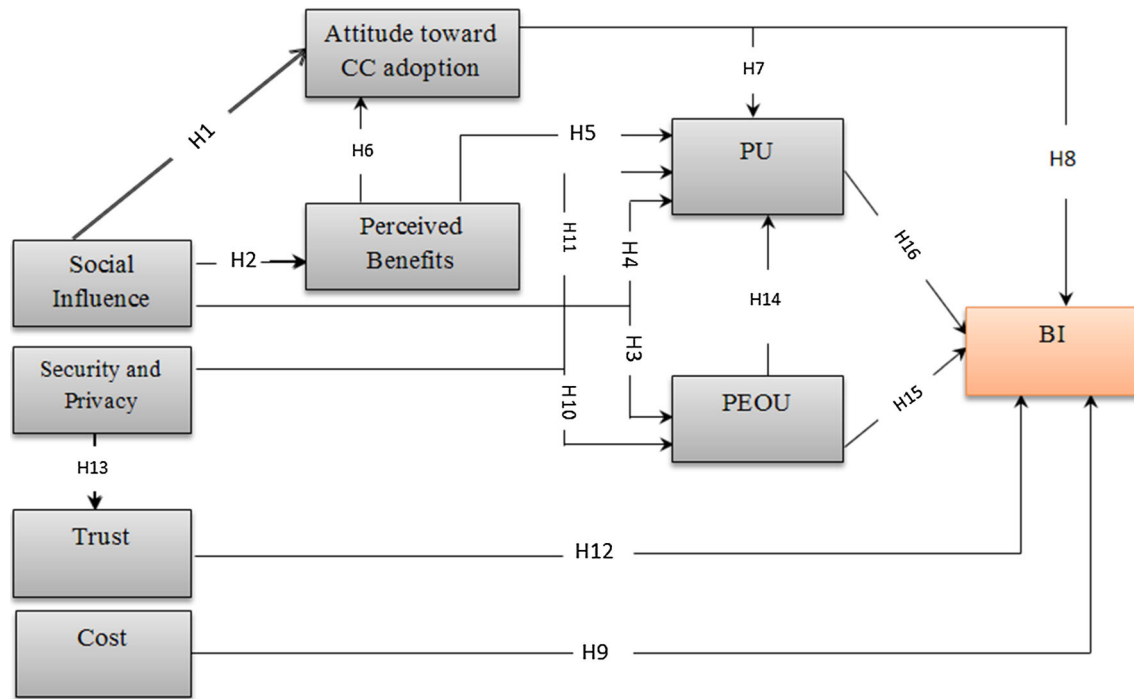


Fig. 1 Cloud based apps adoption model

which individuals are interested in the news about using SaaS and the degree to which they enjoy trying new technologies and services. Furthermore, given that cloud computing generates many potential benefits for banking customers such as reduced waiting times for customers, lower cost, security and privacy concern etc., it is reasonable to infer that the perceived benefits positively influences the users' attitude and intention to adopt cloud computing. High levels of benefit perception toward an object accelerates a consumer's attitudinal orientation and behavior [67]. Empirical evidence from an e-business study reveals that perceived benefit significantly affects e-business adoption [148]. Along the same lines, Beatty et al. [15] suggested that perceived benefit has a positive influence on the corporate website adoption.

According to López-Nicolás et al. [87], PB may have a positive effect on either ATT or PU.

Therefore, we assume the following hypothesis:

H5 PB has a positive effect on PU.

H6 PB has a positive effect on ATT.

“Attitude” is defined as “an individual's positive or negative feelings about performing the target behavior” [3]. Thus, in other words, if an individual has positive feelings about performing a behavior they will also have a strong preference toward performing that behavior, and will thus, be likely to perform that behaviour [13].

In fact, attitude is an important construct when studying individual IT usage [146], and cloud computing literature also indicates that a persons' attitude toward cloud computing adoption has positive influence on that person's behavior intention to use cloud computing [54]. Many researchers have established the importance of the influences of attitude on intentions, as well as confirmed the link between intentions and behavior [38, 60].

Referring to López-Nicolás et al. [87] and Moon and Kim [97], ATT may affect PU and BI. Moreover, individuals with higher levels of personal innovativeness are expected to develop more positive beliefs from new technology [79]. An innovative attitude towards technological innovations may enhance the perception of usefulness [114]. We therefore expected that:

H7 ATT has a positive effect on PU.

H8 ATT has a positive effect on BI.

Cloud computing emphasizes on the delivery of low-cost or free applications anywhere on the Internet. The transaction cost is a significant factor for users when considering whether to use cloud banking services. If consumers perceive the transaction cost of cloud banking service as acceptable, they are likely to adopt it and use it. Thus, a reduction in the transaction costs of the cloud services by the provider will make cloud computing outsourcing more attractive and beneficial. Hamilton [58]

Table 5 Convergent validity

Model construct	Measurement item	Loading	CR ^a	AVE ^b
Attitude toward technology using (ATT)	ATTU1	0.929	0.930	0.877
	ATTU2	0.946		
	ATTU3	0.933		
Behavior intention (BI)	BI1	0.987	0.970	0.917
	BI2	0.946		
	BI3	0.953		
	BI4	0.945		
Cost	COST1	0.961	0.962	0.929
	COST2	0.972		
	COST3	0.959		
Perceived benefits (PB)	PB1	0.967	0.958	0.957
	PB2	0.989		
	PB3	0.990		
	PB4	0.996		
Perceived ease of use (PEOU)	PEOU1	0.959	0.974	0.927
	PEOU2	0.973		
	PEOU3	0.968		
	PEOU4	0.951		
Perceived usefulness (PU)	PU1	0.944	0.938	0.890
	PU2	0.944		
	PU3	0.940		
Security and privacy (S&P)	S&P1	0.990	0.988	0.976
	S&P2	0.987		
	S&P3	0.988		
Social influence (SI)	SI1	0.956	0.958	0.922
	SI2	0.964		
	SI3	0.961		
Trust (T)	T1	0.961	0.977	0.937
	T2	0.974		
	T3	0.975		
	T4	0.962		

^a Composite reliability (CR) = (square of the summation of the factor loadings)/{(square of the summation of the factor loadings) + (square of the summation of the error variances)}

^b Average variance extracted (AVE) = (sum of the square of the factor loadings)/{(sum of the square of the factor Loadings) + (sum of the error variances)}

points that a High transaction cost will negatively impact the intention to adopt cloud computing. In this study we may assume that lower transaction cost has positive effect on intention to adopt cloud computing.

The following hypothesis is then proposed:

H9 cost has a positive effect on BI.

The importance of security and privacy concerns in cloud computing has been broadly discussed and reported in many studies [35, 122, 145]. Chen and Zhao [30] reported that cloud computing security concerns, especially data security and privacy protection issues, remain the

primary inhibitor for adoption of cloud computing services [108]. Based on the consumers' perspective, security concerns remain a major barrier for the adoption of cloud computing. According to a survey from Gartner in 2014, security and privacy remain leading concern for respondents that do not consider public cloud-based models.

Lallmahamood [77] affirms that security and privacy has a significant effect on perceived ease of use when it comes to Internet banking adoption. Also, Charfeddine and Nasri [29] indicated that security and privacy positively influence the perceived usefulness to use internet banking. Therefore, the following hypotheses are introduced:

Table 6 Quality criteria (reliability and convergent validity)

	AVE	Composite reliability	R square	Cronbachs alpha	Communality	Redundancy
ATTU	0.877	0.955	0.614	0.930	0.877	0.276
BI	0.917	0.978	0.806	0.970	0.917	0.278
COST	0.929	0.975		0.962	0.929	
PB	0.957	0.989	0.341	0.985	0.957	0.326
PEOU	0.927	0.981	0.441	0.974	0.927	0.199
PU	0.890	0.960	0.751	0.938	0.890	0.228
S&P	0.976	0.992		0.988	0.976	
SI	0.922	0.973		0.958	0.922	
T	0.937	0.983	0.542	0.977	0.937	0.507

H10 Security and privacy have a positive effect on perceived ease of use.

H11 Security and privacy have a positive effect on perceived usefulness.

The concept of “Trust” has been extensively analysed and a great amount of research has focused on understanding this construct, its acceptable dimensions and its effects on human behavior [17, 49, 86]. Trust in banking is realized as the top important factor for the reason that the financial transactions have to be done online without face-to-face contact. The more a customer trust on cloud computing can lower the perceived online transaction risk and greater the intention to use cloud computing.

In this study, it is assumed that a consumer’s trust in cloud services can provide cloud providers with a set of manageable, strategic levers to build such trust, thereby promoting greater acceptance of cloud computing.

Previous studies consistently show that trust exerts a positive direct effect on intentions [6, 18, 25, 49, 57, 73, 75, 116]. The following hypothesis is therefore developed:

H12 Trust will positively influence behavioral intention to use the cloud

People today rarely keep a large amount of cash with them; they carry plastic cards and transact digitally to minimize loss. As banks continue effectively doing business despite fraud, theft and malpractice, cloud computing should similarly evolve and develop counter-forces to thrive in spite of its security threats. With sufficient trust built with the service providers, customers can store data in the cloud with the same confidence that they keep money and other valuable assets in the banks today [22]. Privacy and security problems are very critical issues which online customers care about. Yoon [143] observed that transaction security had positive effect on customers’ trust and satisfaction. If the corporation relieves customers’ fear of privacy and security, the corporation will win customers’ trust. Huang and Liu’s [61] study found that security and privacy is considered as a key factors affecting trust. Also,

security and privacy has positive influence on trust. Therefore, we hypothesize:

H13 Security and Privacy has positive influence on Trust.

Perceived usefulness (PU) is defined as “the degree to which a person believes that using a particular system would enhance his or her job performance” [39]. Davis also defined perceived ease of use (PEOU) as “the degree to which a person believes that using a particular system would be free of effort” [39].

The TAM model specifies a relationship between PEOU and PU. We expect that this relationship exists in cloud computing as well: when the users experience fewer difficulties in using cloud computing to accomplish tasks, he or she may save efforts in accomplishing the task or accomplish more tasks for the same efforts [38], thereby finding cloud computing to be more useful. Thus, a person’s perceived ease of use of cloud computing will have a positive effect on the person’s perceived usefulness of cloud computing [54].

Extensive research over the past decade provides evidence of the significant effect of perceived ease of use on usage intention, either directly or indirectly [31, 40, 53, 65, 92, 135, 137].

In this study, perceived ease of use refers the users’ perception of the technology as easy to learn and easy to use. The perceived ease of use captures how the consumer perceives the ease of using cloud computing. Consumers also need to have the ability to use the technology based on their competency levels [126]. Thus, given that perceived ease of use is a determinant of consumers’ adoption behavior toward technology innovations, the following hypothesis is proposed PEOU positively influences users’ behavioral intention.

In the context of this research, PU refers to the extent to which bank customers will accept cloud computing if they perceive it to be useful in the workplace or the perception of the consumer that cloud computing will increase job performance. It is helpful for consumers to have a rational

Table 7 Discriminant validity of contracts

	ATT	BI	COST	PB	PEOU	PU	S&P	SI	T
ATT	0.936								
BI	0.797	0.958							
Cost	0.630	0.758	0.964						
PB	0.646	0.684	0.545	0.978					
PEOU	0.759	0.769	0.563	0.689	0.963				
PU	0.769	0.792	0.643	0.707	0.790	0.943			
S&P	0.530	0.623	0.506	0.643	0.527	0.614	0.988		
SI	0.738	0.657	0.514	0.584	0.621	0.717	0.525	0.960	
T	0.645	0.688	0.527	0.665	0.607	0.639	0.736	0.575	0.968

Bold values indicate square root of the AVEs for each construct was greater than the correlations of the construct with the other constructs and satisfied the criterion of the construct level discriminant validity

Values in diagonal represent AVE, while off-diagonal refer to the squared correlation

ATT attitude toward using, BI behavioral intention, COST cost, PB perceive benefits, PEOU perceive ease of use, PRIV privacy, SEC security, SI social influence, PU perceive usefulness, T trust

assessment about the usefulness of an innovation before deciding to use the innovation [42]. Being that cloud computing is still an emerging technology innovation with many potential benefits for consumers, the perceptions consumers have about the technology will affect their behavioral intentions. There is also extensive empirical evidence that supports the significant effect of perceived usefulness on behavioral intention (e.g., [31, 38, 64, 134]). Based on prior research, this study hypothesized the following.

H14 PEOU has a positive effect on PU.

H15 PEOU has a positive effect on BI.

H16 PU has a positive effect on BI.

Thus, the research instrument used in this study was developed using measurement scales identified from previous literature. Changes were made to the selected items to ensure it aligns with the study context. Nine items were constructed and measured. The items include attitude toward using new technology, perceived benefits, social influence, perceived usefulness, perceived ease of use, and behavior intention, all of which are necessary for developing our proposed framework. A list of these constructs, their measurements and definitions are presented in Table 4.

5 Data analysis

The research hypotheses were tested using structural equation modelling based on partial least squares (PLS) method. PLS was proposed by Fornell and Cha [44] as a second generation multivariate technique that has the ability to evaluate the relationships that exist between constructs and their corresponding indicators simultaneously (the measurement model). It can also be used to

minimize the error variance through its structural model [33, 54]. Another major reason why PLS was chosen as technique for analysing data in this study is because of the following reasons stated by [12, 33, 44, 112].

- PLS uses variance technique to predict and explain the variance of a model.
- PLS can work well with a small sample size.
- PLS considers the error in measurement during structural model assessment, without any multivariate normality assumption.

Smart PLS M2 Version 2.0 was used for the data analysis in this study while the bootstrapping method (300 resamples) was employed for computing the level of significance for the path coefficient, weights and loadings, as stated in [33, 54].

5.1 Measurement model

Measurement theory indicates how the constructs are measured. Usually, there are two diverse ways to measure unobservable variables. One approach is referred to as reflective measurement, and the other is a formative measurement. In the reflective measure all indicator items are caused by the same construct, indicators related to a particular construct should be highly correlated with each other. In addition, changes in the constructs lead to changes in the indicators. And, individual items should be interchangeable, and any single item can generally be left out without changing the meaning of the construct, as long as the construct has sufficient reliability.

In contrast, formative measurement models are based on the assumption that the indicators cause the construct. An important characteristic of formative indicators is that they are not interchangeable, as is true with reflective indicators [43, 56]. Therefore, two theories were measured and 33

Table 8 Cross loadings

	ATT	BI	Cost	PEOU	PU	S&P	SI	T	PB
<i>Attitude toward cloud</i>									
Att1	0.929	0.759	0.624	0.705	0.708	0.492	0.693	0.591	0.612
Att2	0.946	0.761	0.625	0.727	0.740	0.523	0.683	0.629	0.597
Att3	0.933	0.719	0.520	0.699	0.712	0.473	0.697	0.593	0.604
<i>Behavior intention</i>									
BI1	0.805	0.987	0.769	0.782	0.808	0.638	0.655	0.694	0.719
BI2	0.723	0.946	0.735	0.702	0.720	0.578	0.606	0.643	0.629
BI3	0.782	0.953	0.6901	0.736	0.757	0.564	0.632	0.637	0.650
BI4	0.742	0.945	0.706	0.724	0.747	0.603	0.621	0.658	0.618
<i>Cost</i>									
Cost1	0.598	0.738	0.961	0.550	0.598	0.471	0.480	0.519	0.515
Cost2	0.631	0.743	0.972	0.545	0.653	0.514	0.517	0.521	0.546
Cost3	0.594	0.710	0.959	0.534	0.608	0.476	0.488	0.482	0.515
<i>Perceived ease of use</i>									
PEOU1	0.719	0.732	0.554	0.959	0.766	0.504	0.608	0.568	0.676
PEOU2	0.741	0.749	0.515	0.973	0.761	0.497	0.610	0.593	0.654
PEOU3	0.733	0.741	0.532	0.968	0.750	0.498	0.594	0.560	0.653
PEOU4	0.726	0.740	0.568	0.951	0.763	0.529	0.580	0.616	0.668
<i>Perceived usefulness</i>									
PU1	0.722	0.756	0.614	0.768	0.944	0.575	0.696	0.628	0.670
PU2	0.720	0.735	0.586	0.744	0.944	0.594	0.663	0.602	0.681
PU3	0.734	0.750	0.619	0.721	0.941	0.569	0.668	0.577	0.649
<i>Security</i>									
SEC1	0.533	0.620	0.494	0.534	0.607	0.990	0.526	0.724	0.647
SEC2	0.522	0.615	0.502	0.510	0.597	0.987	0.510	0.727	0.626
SEC3	0.515	0.610	0.503	0.517	0.616	0.988	0.520	0.730	0.632
<i>Social influence</i>									
Social1	0.687	0.624	0.519	0.607	0.682	0.509	0.956	0.545	0.562
Social2	0.720	0.636	0.479	0.590	0.678	0.504	0.964	0.559	0.570
Social3	0.717	0.632	0.482	0.592	0.705	0.500	0.961	0.552	0.551
<i>Trust</i>									
Trust1	0.614	0.658	0.481	0.591	0.630	0.702	0.572	0.961	0.651
Trust2	0.625	0.670	0.533	0.578	0.604	0.721	0.547	0.974	0.634
Trust3	0.638	0.672	0.535	0.602	0.622	0.702	0.566	0.975	0.636
Trust4	0.622	0.662	0.488	0.580	0.617	0.724	0.541	0.961	0.653
<i>Perceived benefits</i>									
PB1	0.591	0.641	0.519	0.651	0.668	0.624	0.574	0.652	0.967
PB2	0.662	0.701	0.547	0.701	0.702	0.653	0.587	0.682	0.989
PB3	0.666	0.694	0.544	0.702	0.713	0.646	0.596	0.680	0.990
PB4	0.602925	0.636	0.522	0.636	0.682	0.590	0.525	0.585	0.966

The results of cross loading test indicate that the indicator's outer loading for each construct (Bold values) is greater than all of the loading on other constructs. This result confirmed the discriminant validity of the measurement model

constructs were adapted from previous studies. Moreover, based on the Measuring items that all share common theme and are interchangeable. Each construct is operationalized as a reflective model.

The following indicators were used to measure the proposed model:

5.1.1 Convergent validity

The degree of agreement among the various items used to measure the same concept, also known as the convergent validity, was the first to be measured. The average variance, composite reliability and factor loadings were used to

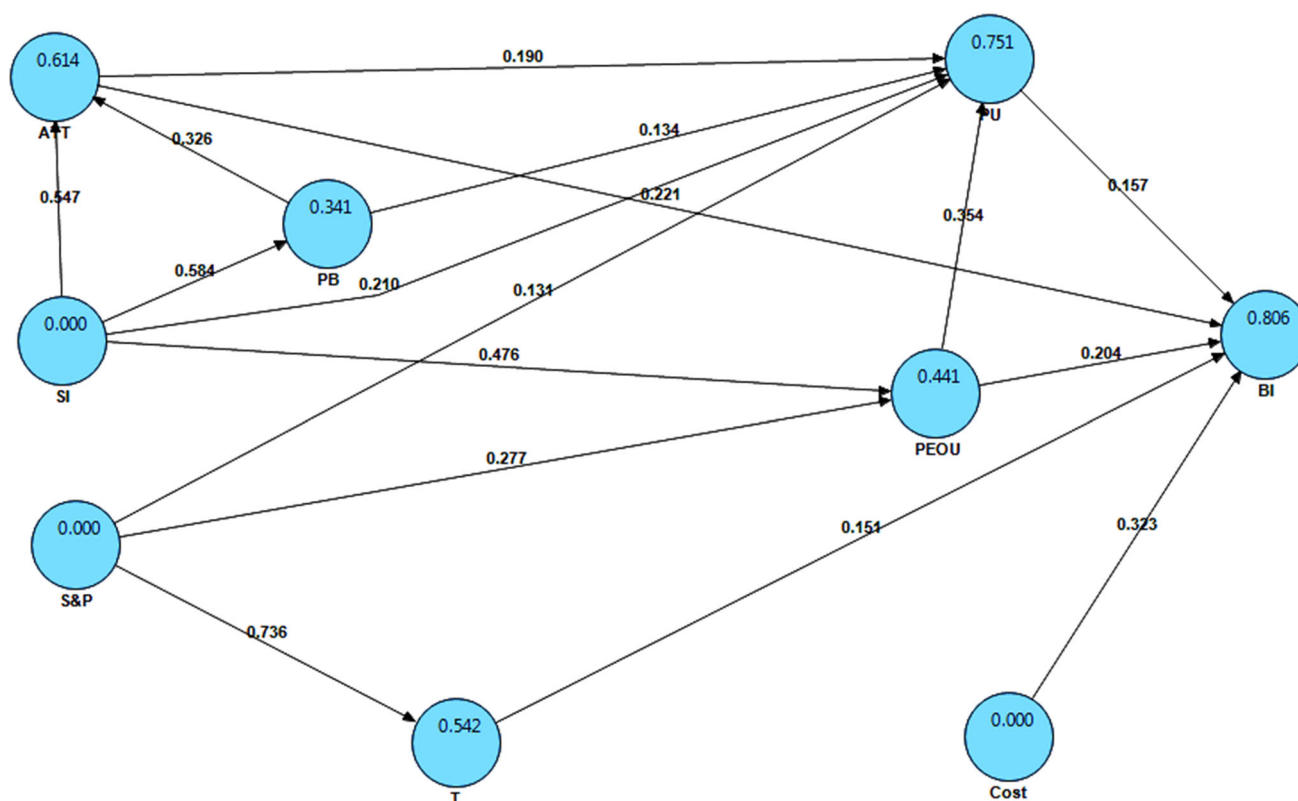


Fig. 2 Results of PLS bootstrapping analysis. *ATT* attitude toward cloud, *BI* behavioral intention to use, *PB* perceived benefits, *PEU* perceived ease of use, *PU* perceived usefulness, *T* trust, *S&P* security and privacy, *Cost* cost; *SI* social influence trust

determine the convergent validity based on the suggestions of Hair et al. [55]. Based on the findings, the recommended value of 6.0 was exceeded for every item loading [32]. Composite reliability values ranged from 0.955 to 0.992, thus exceeding the recommended value of 0.7 (see Table 5), which represents the latent level for the construct indicators [55]. The total amount of variance in the indicators, accounted for by the latent constructs, and represented by the average variance extracted, ranged between 0.877 and 0.976, and exceeded the 0.5 recommended value [55].

For the assessment of the measurement model such as its reflective constructs, two indicators indicators for reliability (Composite Reliability and Cronbach's Alpha) were used [59], while for internal consistency, convergent and discriminant validity were evaluated. The results of the all valuation indicators are presented in Table 6.

5.1.2 Discriminant validity

Discriminant validity is shown when the following two things happen. First, measurement items load more strongly on their assigned construct than on the other constructs and second, when the square root of the average variance extracted (AVE) of each construct is larger than its correlations with the other constructs (Table 7).

As shown in Table 8, all the measurement items loaded considerably stronger on their respective factor than on the other constructs. Table 7 shows the square root of the AVE and the inter-construct correlations. Comparisons of the correlation with the square root of the AVE show that all correlations between the two constructs are less than the square root of the AVE of both constructs [50].

In addition, cross loadings offer another check for discriminant validity. If an indicator has a higher correlation with another latent variable than with respective latent variable, the appropriateness of the model should be reconsidered [43, 56]. As shown in Table 8, all the measurement items loaded considerably stronger on their respective factor than on the other constructs. Table 7 shows the square root of the AVE and the inter-construct correlations. Comparisons of the correlation with the square root of the AVE show that all correlations between the two constructs are less than the square root of the AVE of both constructs.

5.1.2.1 Structural model The casual relationships that exist between the various model constructs are indicated by the structural model [112]. These relationships such as the R value that determines the prediction power of the model, the overall path coefficients estimate, loadings and

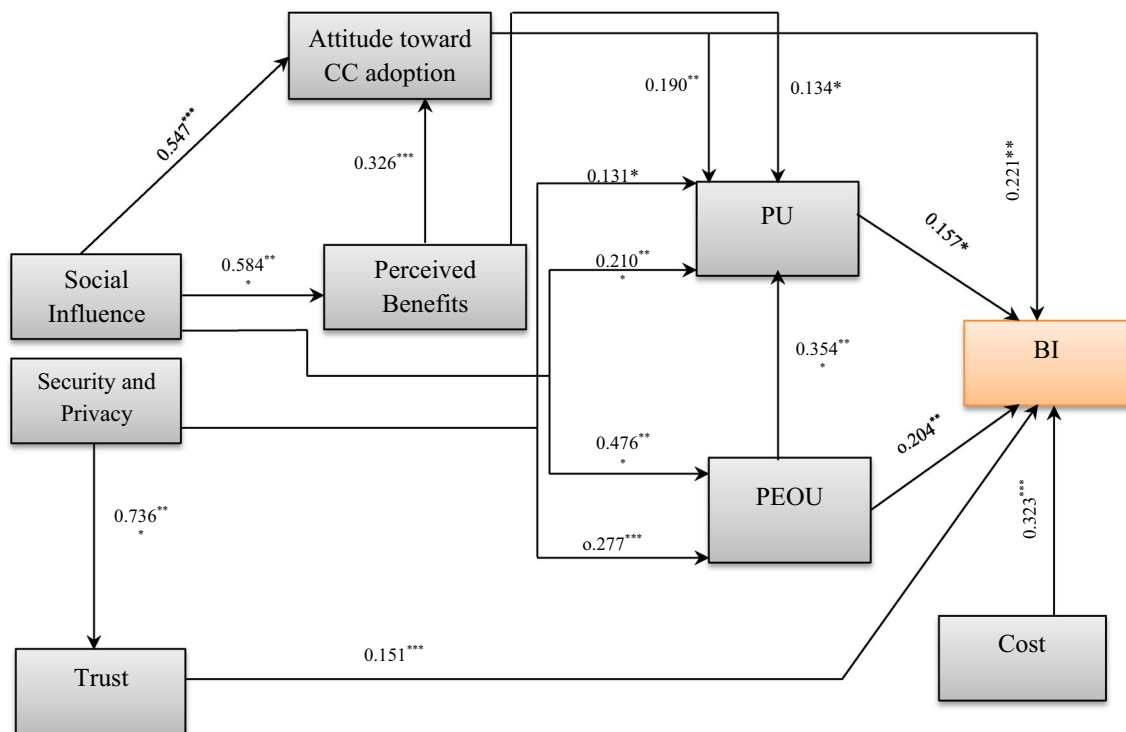


Fig. 3 Significance level of path coefficients among variables. Significance value: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

significance represented by R and path coefficients, together determines the degree of support provided by the data and its ability to hypothesize the model [33, 112]. The structural model results obtained from the output of the PLS is presented in Table 8 and Fig. 2.

After verifying that the outer model estimations requirements were met, an evaluation of the structural model was conducted. The bootstrapping technique was applied to generate 162 samples from 300 cases. This technique was applied with the aim of assessing the structural (or internal) model. According to Henseler et al. [59], the structural model is evaluated taking into account: (a) coefficients of determination of endogenous latent variables; (b) path coefficients (in terms of sign, magnitude and significance) and; (c) Cohen effect-sizes on the endogenous variables at the structural level. Given the previously mentioned aspects, the model depicted in Fig. 2 presents the PLS model results.

5.2 Path coefficients and predictive ability

Significant item loadings were exhibited in Table 8, while Table 9 only shows that path coefficients were significant between constructs in the model. Figure 2 demonstrates the significant relations between constructs. It is clear that the path coefficient “S&P → T” with 0.736 had the greatest positive direct effect, followed by “SI → BP” with 0.584.

As for the R^2 value of endogenous constructs, “PU” exhibited the best explanatory ability in the model with a score of 75.1%. Overall, the combination of “T”, “SI”, “S&P”, “PB”, “ATT”, “PU”, and “PEOU”, had the predictive ability for 80.6% of the “BI”, implying that the confidence ability of the entire model in terms of explanatory power and estimation of the predictive power of the independent variables. We conclude therefore that the model provides an adequate predictive power for the behavioral intention to use cloud computing by considering the independent variables, including social influence, security and privacy, cost. The second key criterion for evaluating the PLS structural equation model is an assessment of the path coefficients, which indicates the strength of the relationship between the latent variables of the research model. This value should be significant, and at least at the 0.05 level [132]. The evaluation test results of the path coefficient and t-value for the PLS structural model is presented in Table 9. The t-value were computed using 300 re-sample with bootstrapping [33, 69]. This method is compatible with previous research on technology acceptance [113]. The results of bootstrapping are shown in Fig. 2. The significant structural relationships among the research variables and path coefficients are presented in Table 9, as well as in Fig. 2. The results indicate that all path coefficients were positive and significant. The path coefficients (beta) of H1–H16 are positive and all are supported (see Table 8).

Table 9 Overall analysis

Hypotheses	Path coefficient (β)	t-value	Support
H1: Social influence -> attitude toward cloud Computing ATT	0.547	11.00***	Yes
H2: Social influence -> perceived benefit	0.584	10.49***	Yes
H3: Social influence -> perceived ease of use	0.476	8.16***	Yes
H4: Social influence -> perceived usefulness	0.210	3.86***	Yes
H5: Perceived benefit -> perceived usefulness	0.134	2.08*	Yes
H6: Perceived benefit -> Attitude toward cloud Computing	0.326	6.17***	Yes
H7: Attitude toward cloud computing -> perceived Usefulness	0.190	2.67**	Yes
H8: Attitude toward cloud computing -> behavior Intention	0.221	2.90**	Yes
H9: Cost -> behavior intention	0.323	5.22***	Yes
H10: Security and privacy -> perceived ease of use	0.277	4.35***	Yes
H11: Security and privacy -> perceived usefulness	0.131	2.19*	Yes
H12: Trust -> behavior Intention	0.151	3.15***	Yes
H13: Security and privacy -> trust	0.736	19.83***	Yes
H14: Perceived ease of use -> perceived usefulness	0.354	4.79***	Yes
H15: Perceived ease of use -> behavior intention	0.204	2.98**	Yes
H16: Perceived usefulness -> behavior Intention	0.157	1.97*	Yes

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Fig 3 summarizes the PLS structural analysis results. In line with the results found in the extant literature listed.

6 Discussion

New technology adoptions are usually influenced by a number of factors, which may range from technology issues to the characteristics and attitudes of users. The complexity and importance attached to the attempt to explain the causes or reasons for users' adoption or rejection of new technologies have led to the postulation of many theories and development of several models.

The simplicity and understandability of TAM has placed it as one of the most widely accepted and used models, regarded among the best models for adoption studies in IS domain [107]. The development of TAM-diffusion theory model (TAM-DTM) by López-Nicolás et al. [87], paved way for construction of a novel research framework dissimilar to traditional earlier modified versions of TAM. Eight constructs were used in TAM-DTM model to explain the influential factors that determine the adoption of cloud computing. One major reason for the choice of using TAM-DTM model in this study is because of its suitability and compactness for understanding the factors that influence cloud computing adoption. More so, its added imperative constructs like privacy, security, trust and cost, gives it a superior advantage over other conventional TAM models. A t test measure was used to determine the influence of each construct. Based on the findings, the results suggest that there is a positive significant influence on four major

constructs by social influence. The constructs are: ATT ($t = 11$, $p < 0.001$), PB ($t = 10.49$, $p < 0.001$), PU ($t = 3.86$, $p < 0.001$) and PEOU ($t = 8.16$, $p < 0.001$). The findings are in agreement with those of López-Nicolás et al. [87] and Wu [139]. Another major finding based on the results of this study indicate that behavioral intention is positively influenced by customers' attitude towards the use of cloud computing services/applications ($t = .29$, $p < 0.01$), and is in agreement with the assertions of Alotaibi [6], López-Nicolás [87] and Wu [139]. Also, a significant positive influence on perceived usefulness by the perceived benefits was observed from the results ($t = 2.08$, $p < 0.05$) and is supported by the findings of some previous studies [87, 139]. This implies that PU determines the degree to which cloud computing is perceived by customers as a more efficient, faster and convenient way of accomplishing tasks. The findings of the study that some of the biggest issues in cloud computing adoption, implementation and use are privacy, security and trust issues are also supported in literature [2, 8]. Based on the results of this study, privacy and security were shown to have a significant positive influence on PEOU ($t = 4.35$, $p < 0.001$) and are consistent with the findings of Lallmahamood [77]. Also, security and privacy were found to have a positive influence on PU ($t = 2.19$, $p < 0.05$) and is in agreement with the findings of Charfeddine and Nasri [29] and Wu [139]. On a similar note, security and privacy were shown to have a major positive influence on trust ($t = 19.83$, $p < 0.001$). Huang and Liu [61] had earlier shown that security and privacy are very important factors that influence buyers' trust of cloud platform providers.

The results of this study shows trust as a major factor that positively affects behavior intention ($t = 3.15$, $p < 0.001$). Several studies, including Gangwar et al. [47], López-Nicolás et al. [87] and Wu [139] have shown the influence of PEOU on both BI and PU. The results of this study shows that PEOU has a positive significant influence on PU ($t = 4.79$, $p < 0.001$) as well as a positive significant influence on BI ($t = 2.98$, $p < 0.01$). Previous studies of Benlian and Hess [19], Garrison [48] and Oliveira et al. [100] all agree with this assertion, implying that cost saving is a major driver of cloud-based solutions adoption in several sectors like education, services, logistics, finance and manufacturing. The above assertion is also in agreement with the findings of this study, which suggests that cost has a positive effect on behavior intention ($t = 5.22$, $p < 0.001$) and is the main driver for the adoption of cloud computing. Thus, the cost of transaction strongly influences customers use of cloud computing services, such that the lower the cost, the more the likelihood of adoption. López-Nicolás et al. [87] posit that PB may have a positive effect on either ATT, which also is asserted by the findings of this study that PB has a positive and significant influence on ATT ($t = 6.17$, $p < 0.001$), which consequently has a positive effect on PU ($t = 2.67$, $p < 0.01$). This result is in agreement with that of López-Nicolás et al. [87] thus implying that PU is influenced by ATT. Furthermore, the results show that ATT had the highest explanatory power (61.4%), followed by PB (34.1%), T (54.2%), PU (75.1%), PEOU (44.1%).

In summary, it can be deduced from the above findings that factors like trust, cost, security and privacy are very influential in cloud computing adoption. This goes to corroborate the hypothesis of this study. Also, BI can be said to be directly influenced by perceived usefulness, and is in agreement with the findings of King and He [74] and Wu [139]. In addition, based on the important items measured (see Fig. 2), the following possible explanations are proposed: (1) the key driver for cloud computing adoption based on users' perspective is cost; (2) the adoption of cloud computing under security and privacy can be said to be secure and trustworthy.

The results of our study indicate that, for the purpose of critical factors for adoption of cloud computing in banking sector, this paper has an integrated and developed model that incorporates TAM and DOI related theories with additional critical constructs such as Cost, security, Privacy and trust. The results of this study indicate that all hypotheses were supported by the data. From these significant and supported hypotheses, we can derive several implications as follows: (1) SI has been shown to have positively wide-ranging influences on ATI, PB, PU, PEOU; (2) S&P, had positive effects on T; (3) ATT&T& Cost had positive effects BI; (4) PEOU had a positive influence on

PU; (5) PU & PEOU had a positive influence on BI; (6) S&P had a positive effect on PU & PEOU. Additionally, the results of this study provide support for the research model presented in Fig. 2 the construct with the highest explanatory power was BI (80%), followed by PU (75%), ATT (61%), T (54%), PEOU (44%), and PB (34%). This finding is consistent with several prior studies that have identified ATT and PU as two key determinants of BI [128]. The extended TAM with DOI model is capable of explaining a relatively high proportion of variation of intention to adopt cloud computing. Hence, there is need for clients to give trust to the cloud providers in a similar way they would trust banks to safeguard their money. And lastly, cloud providers must earn the trust of users by proving to be reliable and trustworthy. Thus, it is the contention of this paper that customers must yield the same degree of trust they offer banks to cloud providers if a widespread adoption of cloud computing is to be witnessed soon.

7 Conclusion

Cloud computing has taken a leading role in today's information system technologies with such attractive properties as scalability, speed, pay as you go pattern and cost reduction. Cloud computing has emerged as a topical field in the research community because of its transformational power on various platforms, infrastructure as a service and computer software. Despite its numerous benefits and advantages, cloud computing suffers a low user adoption rate coupled with the fact that there is a dearth of literature in user studies irrespective of the crucial role users play in the success and failure of emerging technologies. This paper presented a study aimed at investigating users' perspective for the adoption of cloud computing in Malaysia. As a baseline, it utilizes the TAM-DTM model together with three additional factors that influence cloud computing adoption in the region so as to achieve the study goals. Factors such as cost, trust, security and privacy were identified as the major drivers of cloud computing adoption in the banking sector. The study offers a meaningful base for cloud computing providers to understand the deeper issues surrounding user adoption of cloud computing particularly, in the banking sector. Some limitations of this study include the small sample size used, which may limit the statistical significance of this study.

Therefore, in order to answer the main research question in this study, theory is selected as basic theory and new factors are added to this theory to develop a comprehensive research model. A set of questionnaire is designed based on developed research model and distributed among respondents. Finally, results of the PLS analysis show that

attitude, behavior intention, cost, perceived benefit, perceived ease of use, perceived usefulness, security and privacy are the main factors which influence the adoption of cloud computing by bank customers.

8 Contributions and implications

The study makes an important contribution to the body of the knowledge, particularly in the domain of new technology adoption. Its major theoretical contribution emanates from the use of a TAM-DTM to address the knowledge gap in customers' perspectives for the adoption of cloud computing in the banking sector, no existing literature has before now, undertaken a comprehensive study on the factors that influence the cloud computing adoption from the customers' perspective. The instrument used in this study has been verified for reliability, validity, and discriminant test. It is therefore fair to conclude that the research model and the instruments provide a sound basis for understanding these factors and which can be used as a foundation for future innovative studies by other researchers.

The practical implications of this study, however, hinges on results from the study, which can be by both the banking industry and cloud computing providers to narrow their judgements based on a better understanding of the factors that influence customers' perspectives on the adoption of cloud computing in the banking sector. This can help cloud providers prioritize their services after based on customer's judgements. It can also help them gain better insight into customers' preferences, thereby focusing on better customer relations, human relations and financial management, all of which can help the banks to retain not just the existing customers but to attract new ones. This study, therefore, attempts to offer valuable information to the banking sector and organizations as a whole, that may come handy in decision making regarding cloud computing adoption. Lastly, it offers cloud providers a first hand knowledge of the most significant and critical factors that must be considered in the provision of cloud-based applications and services to consumers.

9 Limitations and further research issues

Like other studies, this research is not without limitations. One major limitation of this study is that the sample population used for the study is limited to the Malaysia residents. This is because the study was conducted primarily in Malaysia. This may have some influence on the study results, although the findings were validated with related studies from other countries. A second limitation of this

study is that the adopted model used for the study, TAM-DTM, may not have explicitly pointed out the major constructs and variables in the proposed model. However, in order to address this limitation, future studies may wish to consider the use of more than one theoretical model when investigating complex situations like customers' perspectives of cloud computing adoption in order to better express the understanding of the IT adoption process.

Other limitations include the small sample size used, which may limit the statistical significance of the study. The study, therefore, recommends the testing of the proposed model with a larger sample size as an area for future studies. Nevertheless, the proposed model in this study should deliver a sound basis for future studies in this domain.

Compliance with ethical standards

Conflict of interest This paper received no funding from any third party and neither of the authors has any conflict of interest.

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